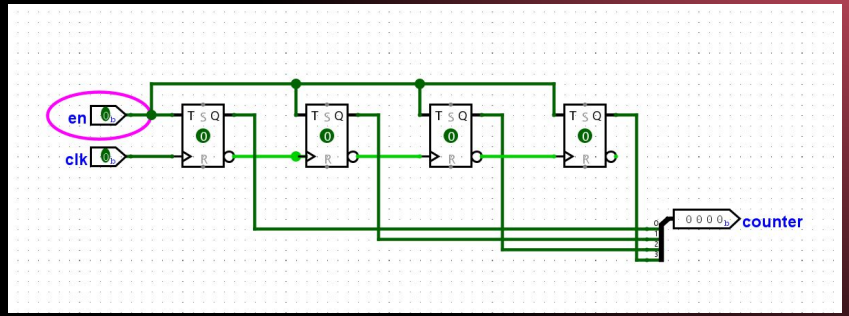
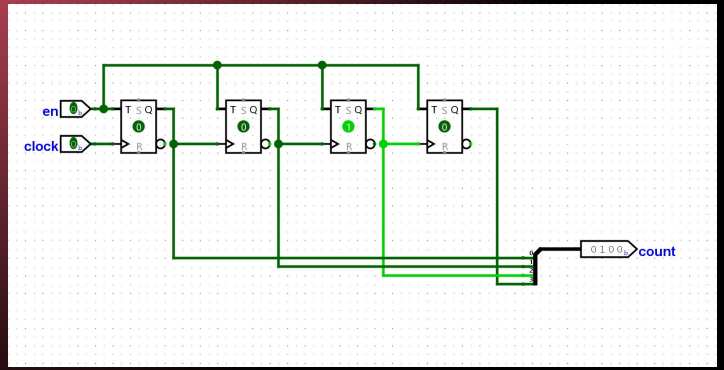
Counters

Chelsea Sisneros

Counters



This is just the up counter, it counts up from 0 to 15.

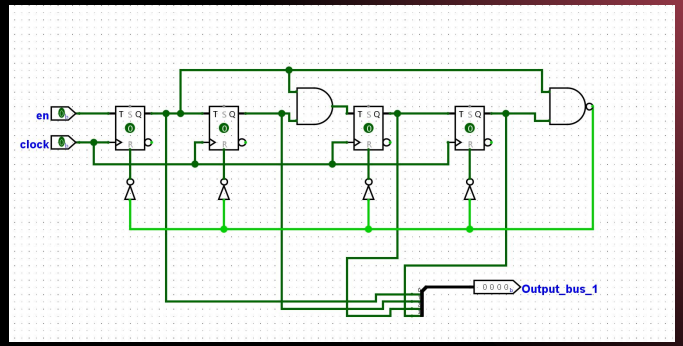


This is just the up counter, it counts down from 15 to 0.

There is a minor difference between these two, the down counter connects the clock to Q, while the up counter connects each clock directly.

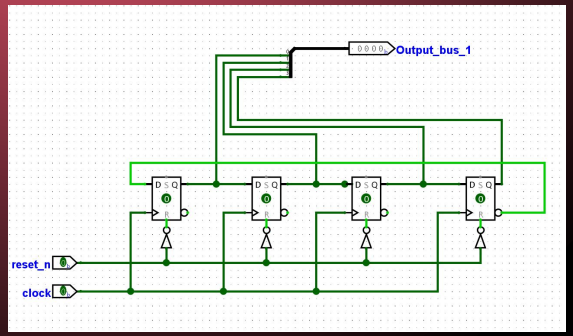
More Circuits/Flip-Flops

Image 1



This is the Mod-5 and this circuit counts up from 0-4.

Image 2



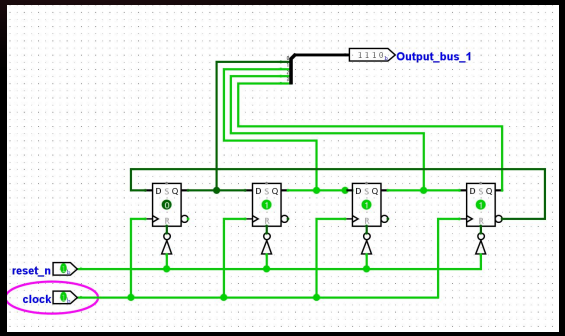
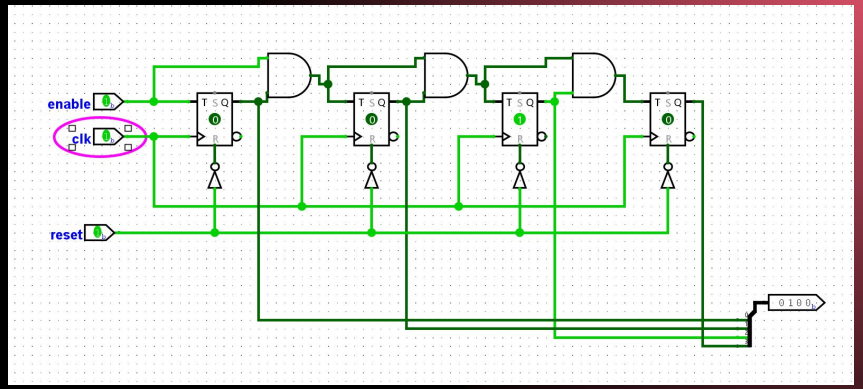


Image 3

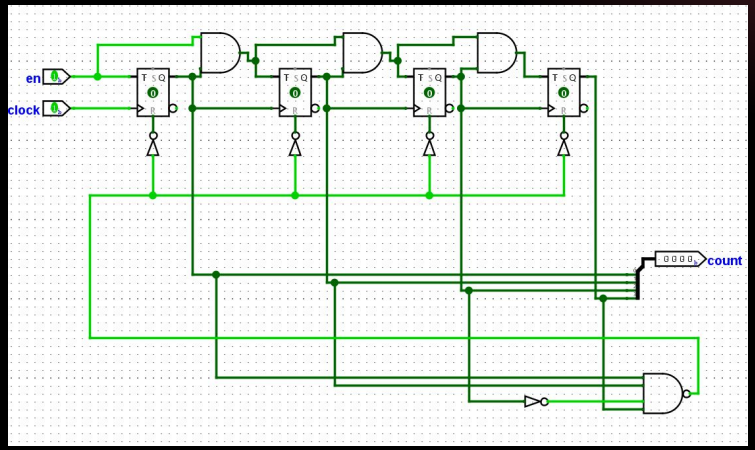
This is the Johnson circuit and this counts up from 0 to 15 and it also has a reset button. Image 2 shows the clock at 0 and Image 3 shows the clock before being rest.

In order to reset this, you click the resent_n and it will revert it back to 0.

Counter Enable with Reset Button

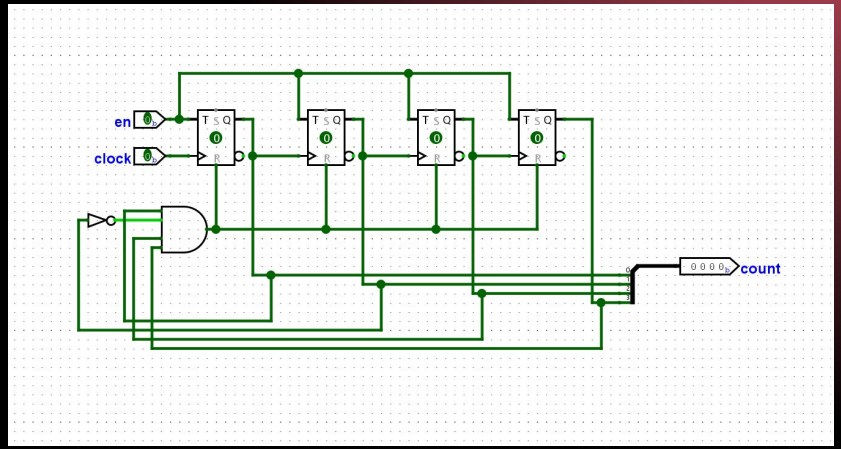


This circuit uses T-flip-flops and has a reset button, this counts up from 0-15.



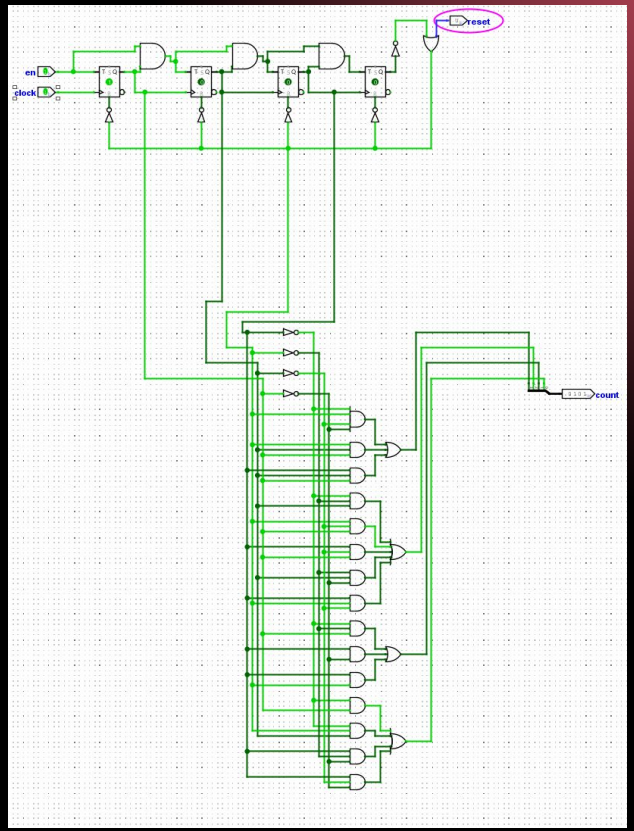
After adding more logic to this, it was able to count from 0 to 10 because the added logic counts up to 11 and resets it back to 10.

Counter Reset pt. 1



This is the Counter Reset after I rearranged the previous logic gates. This model only flipped from 15 to 14 and I lost ideas on how to fix it.

Counter Reset pt. 2

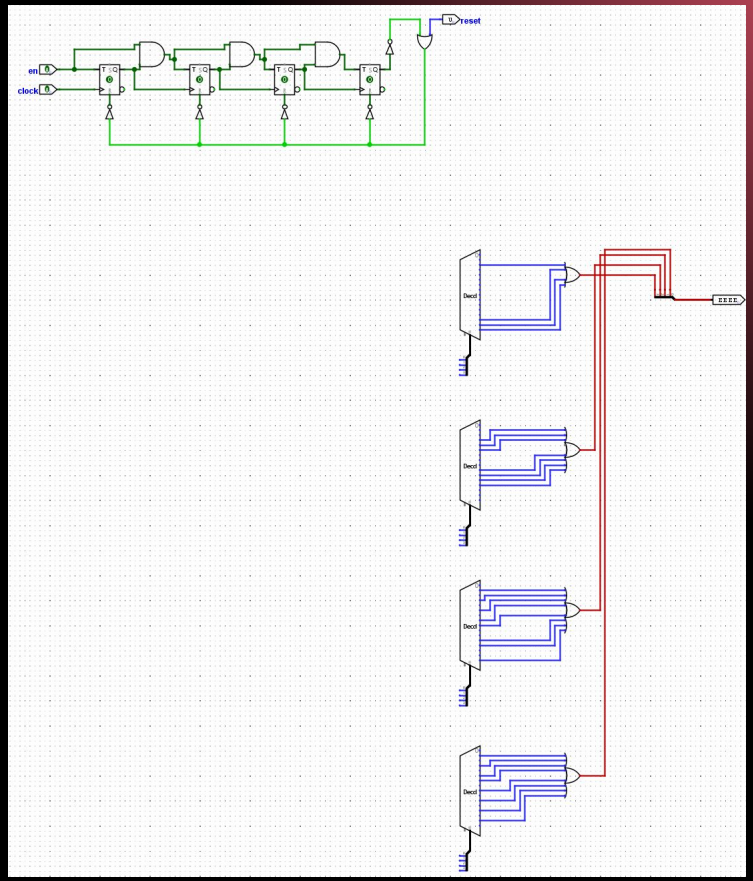


I decided to try to use the truth table and integrate it with the pieces from the counter enable. I tried to use the counter reset, which worked from getting the counter to go through 0 to 10 and then reset. I encountered

This gave me issues I had a hard time getting it to count down. I realized I needed more pieces to actually get the down counter to understand we're starting from 10.

This specific model didn't work, it flipped from a couple random numbers and the reset button didn't work.

Counter reset pt. 3



This is the last thing I tried, i thought using decoders would be easier since it's more uniform, i got some assistance from a friend, but I couldn't figure out how to connect these directly to the logic on top.